

Evaluation of real-time performance of local-level flu forecasts for the 2025-2026 season in a subset of locations in the United States

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Our Goal

We want to **systematically evaluate forecasts** generated at the local level to answer the following questions:

- ① How does **local level forecast performance compare to the performance of aggregated forecasts** within and across different models, and how do **characteristics of the locality and the aggregate population** impact the relative value of local forecasting?
- ② Which **models/methods perform best** for both local and aggregate locations, accounting for **variables that impact a model's overall forecast performance**?

Analysis Roadmap

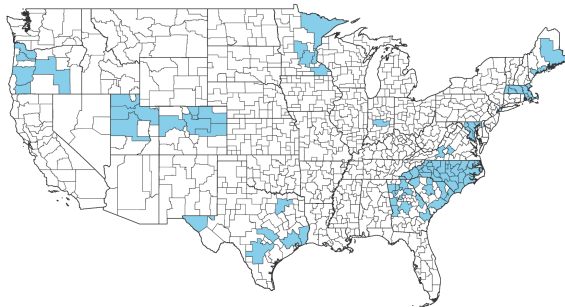
	Model Performance	State vs. Local
Forecast Metrics	Weighted Interval Score (WIS), 95% Coverage <i>[Slides 7-8]</i>	Relative WIS, Scaled Relative Skill Scores <i>[Slides 9-12]</i>
Regression	Random-Effects GAM <i>[Slides 17-19]</i>	Random-Effects GAM with Offset <i>[Slides 13-16]</i>

Flu MetroCast Hub

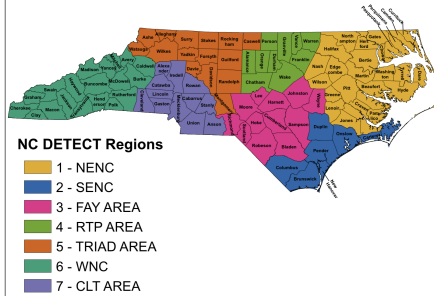
While other forecasting Hubs such as the FluSight Challenge or COVID-19 Forecast Hub are often based at the state and national level, the **Flu MetroCast Hub is designed to focus on the local level**, making it an excellent resource to evaluate.

- The Hub solicited **HSA (Health Service Area) or metro-area level predictions and predictions for the corresponding aggregate location** (usually the state).
- The target was the **percent of Emergency Department (ED) visits due to flu within an epi-week** and the Hub predictions were in horizons ranging from 0 to 3 weeks (with week -1 being an optional horizon), yet only **scored on horizons 0 to 3**.
- The local jurisdictions were chosen from within the aggregate jurisdiction based on **conversations with public health partners at either the local or jurisdiction-level**, who identified which locations would be of **greatest public health relevance**. Additional considerations were made to ensure **population size was deemed sufficiently large** (> 250,000) for forecasting, though this is not true in all cases.

Hub Jurisdictions



NC DETECT Regions



Jurisdiction

56 HSAs + 12 States
7 NC Regions + State
NYC

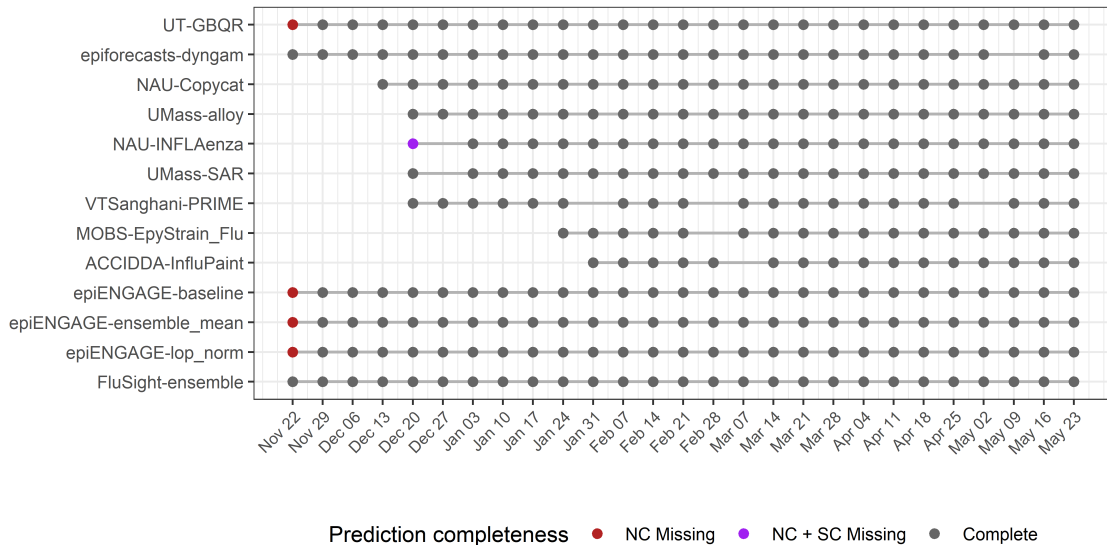
Data Source

CDC NSSP
NC DHHS
NYC DOHMH

Target Outcome

Flu % of ED visits
Flu % of ED visits
ILI % of ED visits

Summary of Model Submissions



Weighted Interval Score (WIS) and Relative WIS (rWIS)

The **interval score** penalizes **dispersion** ($u_\alpha - l_\alpha$), **underprediction** ($y < l_\alpha$), and **overprediction** ($y > u_\alpha$).

$$IS_\alpha = (u_\alpha - l_\alpha) + \frac{2}{\alpha}(l_\alpha - y)\mathbf{1}_{(y < l_\alpha)}(y) + \frac{2}{\alpha}(y - u_\alpha)\mathbf{1}_{(y > u_\alpha)}(y) \quad (1)$$

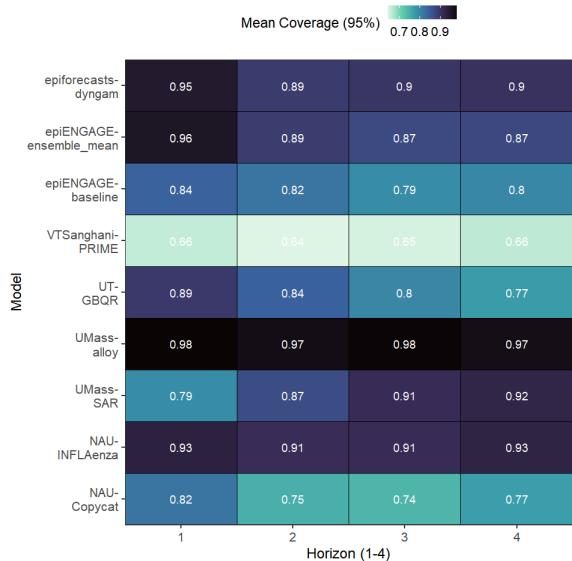
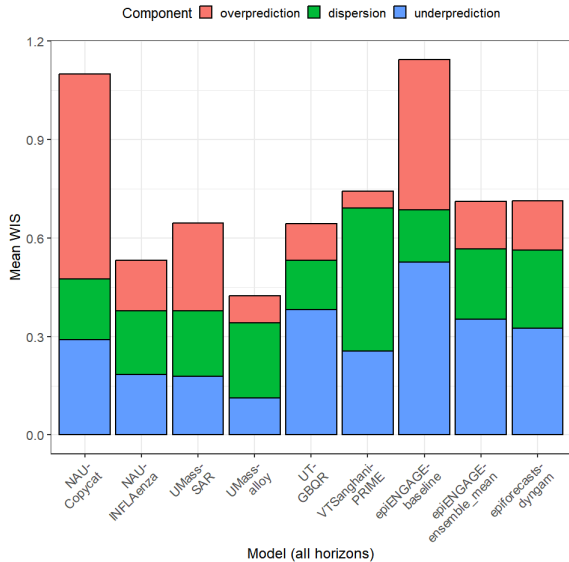
The **weighted interval score** (WIS) is a **weighted average** of the interval scores over K prediction intervals and the absolute error of the median forecast m , with weights $w_k = \alpha_k/2$.

$$WIS = \frac{\frac{1}{2}|y - m| + \sum_{k=1}^K \frac{\alpha_k}{2} IS_{\alpha_k}}{K + \frac{1}{2}} \quad (2)$$

The **relative weighted interval score** (rWIS) is computed as the **ratio** of a model's weighted interval score to the baseline weighted interval score. Since lower WIS is better, rWIS of lower than 1 means the **model performs better than the baseline and vice versa**.

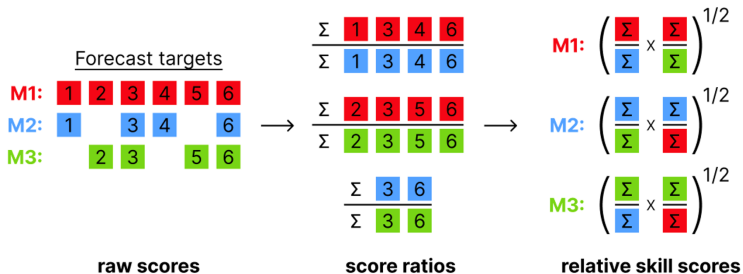
$$rWIS = \frac{WIS_{\text{model}}}{WIS_{\text{baseline}}} \quad (3)$$

WIS component breakdown and 95% coverage per model



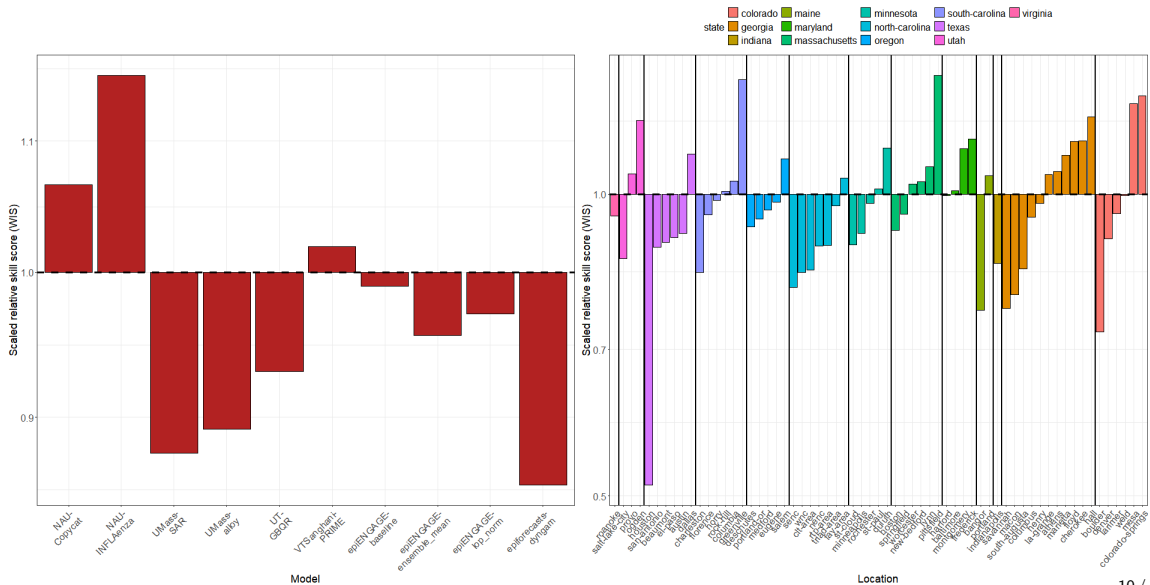
Relative Skill Scores (RSS) and Scaled RSS (SRSS)

A mean score ratio is calculated for every pair of models using only **overlapping forecasts**. The **relative skill score** of a model is the **geometric mean** of all pairwise score ratios involving that model. If a baseline model is available, scores can be **scaled** relative to the baseline. If scaled, then the same principle from rWIS applies where SRSS of lower than 1 means the **model performs better than the baseline and vice versa**.



Credit: scoringutils documentation

SRSS of models and locations

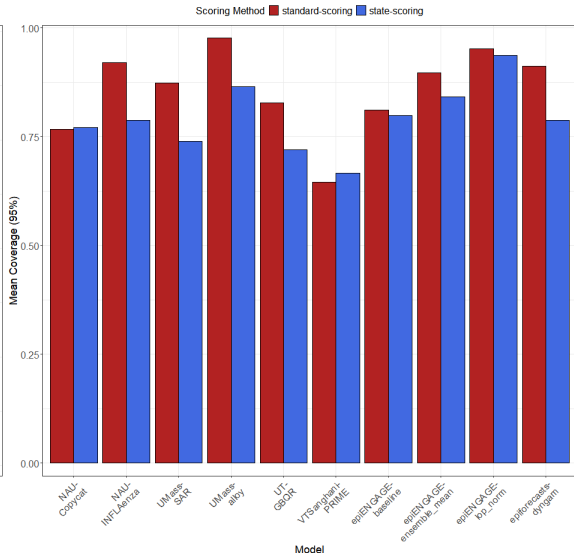
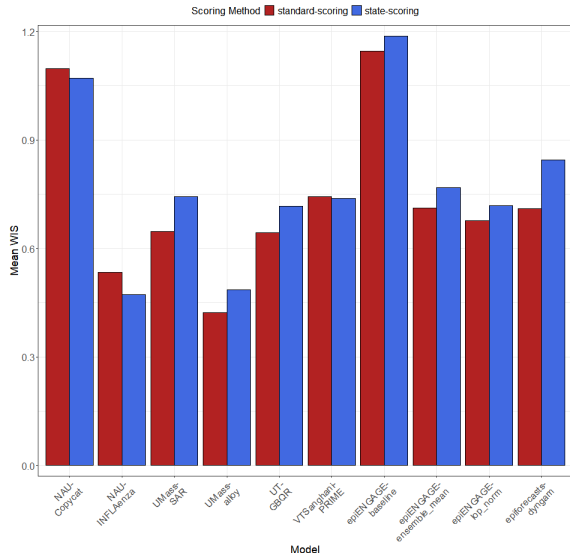


Forecast scoring types

A **standard-scoring forecast** are predictions made by a model on a local (or aggregate) jurisdiction informed by previous observed data of that local (or aggregate) jurisdiction. Informally, these will be referred to as **local-level forecasts**.

A **state-scoring forecast** (on a local jurisdiction) are predictions made by a model on a state (aggregate) jurisdiction informed by jurisdiction observed data of that state (aggregate) jurisdiction. Informally, these will be referred to as **state-level forecasts**.

Standard-scoring vs. state-scoring model performance



Model-based evaluation of aggregate (state) vs. local forecast performance

We first assume **independent and identically distributed errors** on the **log-transformed WIS scores** from the predictions and observations.

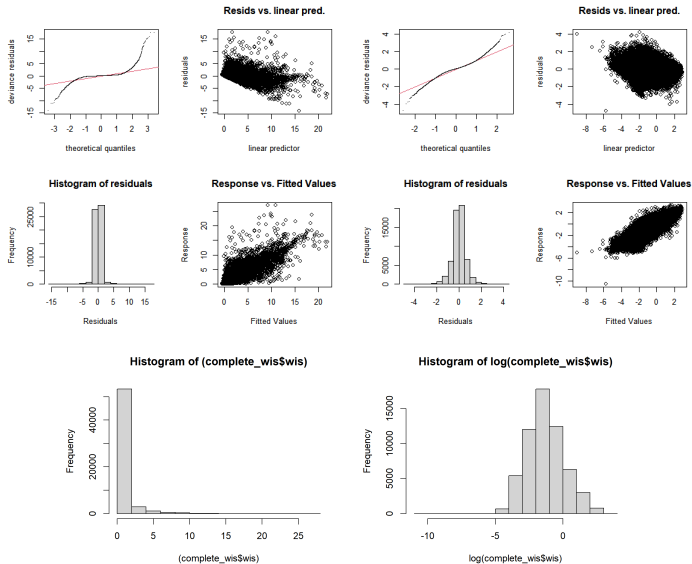
$$\log(\text{WIS}_{h,d,l,m}^{\text{local}}) \sim \text{Normal}(\mu_{h,d,l,m}^{\text{local}}, \sigma) \quad (4)$$

We then model the **expected WIS** of a particular forecast horizon h on forecast date d at location l for model m with a generalized linear model.

$$\mu_{h,d,l,m}^{\text{local}} = \beta_0 + \text{WIS}_{h,d,l,m}^{\text{aggregate}} + \beta_{1,m} + \beta_{2,l} + \beta_{3,h} \quad (5)$$

where h is the forecast horizon, d is the forecast date, l is the local location of the forecast (the borough or metro area), and m is the model. Location, horizon, and model are modeled as random effects. The $\text{WIS}_{h,d,l,m}^{\text{aggregate}}$ **offset should roughly account for the average forecast difficulty** of a particular forecast location, forecast date, and horizon for the particular model.

Why log-transform?



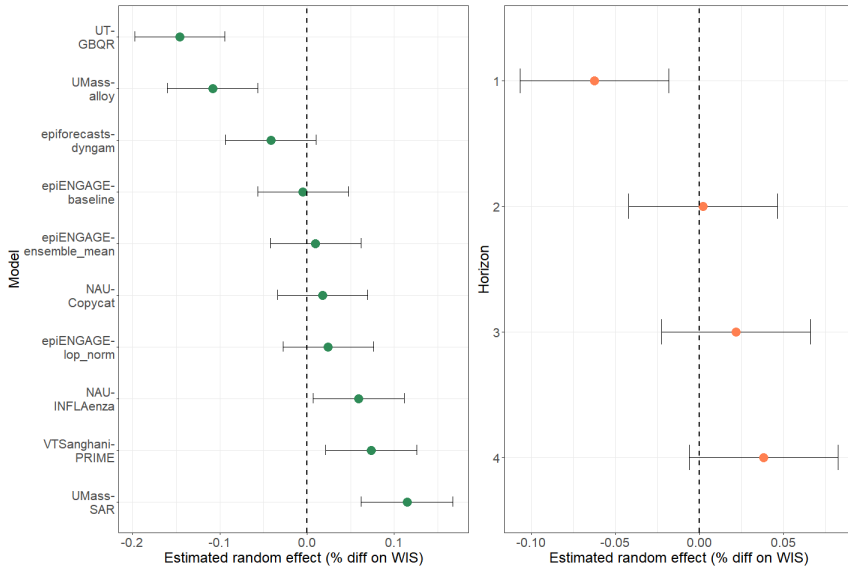
Local vs. aggregate formula in R

This model **log-transforms** both WIS and state WIS (state-scoring WIS) where the **offset** functions as a way to **understand forecast difficulty relative to each specific combination** of date, horizon, location, and model to understand if local does substantially better.

Listing 1. Generalized additive model (GAM) formula with R package `mgcv`

```
model <- gam(  
  log(wis) ~ # log-transformed WIS  
    offset(log(state_wis)) + # log-transformed state_wis offset  
    s(model_id, bs = "re") + # random effect of 'model_id'  
    s(location, bs = "re") + # random effect of 'location'  
    s(horizon, bs = "re"), # random effect of 'horizon'  
  family = gaussian(), # Gaussian distribution assumption  
  data = complete_wis_filtered, # data source  
  method = "REML" # method with superior stability compared to GCV  
)
```

Some estimated random effects with the offset



Model-based evaluation of model performance

Once again, we will assume we have **independent and identically distributed error** around the **log-transformed WIS scores** on the data.

$$\text{WIS}_{h,d,l,m} \sim \text{Normal}(\mu_{h,d,l,m}, \sigma) \quad (6)$$

We will model the **expected WIS** of a particular forecast horizon h on forecast date d at location l for model m with a generalized additive model which **incorporates predictors of forecast performance in potentially non-linear ways**.

$$\mu_{h,d,l,m}^{\text{local}} = \beta_0 + \beta_{1,m} + \beta_{2,l} + \beta_{3,h}, \quad (7)$$

where h is the forecast horizon, d is the forecast date, l is the location of the forecast (here could be either local or aggregate), and m is the model. Model, location, and horizon are all modeled as **random effects on the log-transformed scores**.

Model performance formula in R

This model log-transforms WIS and then treats the rest of the predictors as **random effects** centered at 0 **that represent deviation from the mean** for each predictor's value.

Listing 2. Generalized additive model (GAM) formula with R package mgcv

```
model <- gam(  
  log(wis) ~ # log-transformed WIS  
    s(model_id, bs = "re") + # random effect of 'model_id'  
    s(location, bs = "re") + # random effect of 'location'  
    s(horizon, bs = "re"), # random effect of 'horizon'  
  family = gaussian(), # Gaussian distribution assumption  
  data = standard_wis , # data source  
  method = "REML" # method with superior stability compared to GCV  
)
```

Some more estimated random effects

